

TED(10)4115

Reg.no.....

(REVISION 2010)

Signature.....

Fifth Semester Diploma Examination in Wood and Paper Technology

PROCESS CONTROL AND INSTRUMENTATION

(MODEL QUESTION PAPER)

- I Answer the following questions in one or two sentences (Each carries 2 marks) 5x2=10

PART-A

- 1 State the term Instrumentation.
- 2 List out any two temperature measuring Instrument
- 3 Define Pyrometer
- 4 Define the term consistency
- 5 State control valves

- II (Answer any Five of the following) 5x6=30

PART -B

- 1 Explain about the Range of an Instrument
- 2 Classify the flow measuring Instrument
- 3 Draw and describe about vacuum gauge
- 4 Explain in detail about the freeness measurement
- 5 Describe about finding the thickness of a paper
- 6 Briefly describe the modern process control system
- 7 State P,PI and PID control in process control system

(Answer one full question from each module)

4x15=60

PART-C

MODULE I

- III a Explain the application of instrument in wood and paper industry 8
- b Describe the performance characteristics of an instrument 7
- OR
- IV a Summarize the various factors to be considered in the selection of instrument 8
- b Distinguish between the transducers and transmitters 7

MODULE II

- | | | | |
|---|---|---|---|
| V | a | Evaluate the pressure measurement using | |
| | | (i) Bourdon Tube Pressure Guage | 8 |
| | | (ii) Capacitance Type Pressure Guage | 7 |

OR

- | | | | |
|----|---|---|---|
| VI | a | Differentiate between Orifice meter and Ventury meter | 8 |
| | b | Describe about pyrometer | 7 |

MODULE III

- | | | | |
|-----|---|--|---|
| VII | a | Explain in detail the construction and operation of any one instrument | |
| | | (i) Humidity measurement | 8 |
| | | (ii) PH measurement | 7 |

OR

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|------|---|--|---|
| VIII | a | Describe with a neat sketch for detecting consistency of a paper | 8 |
| | b | Explain two process control variables in pulp ,paper and wood industries | 7 |

MODULE IV

- | | | | |
|----|---|-------------------------|---|
| IX | a | Write short notes on | 5 |
| | | (i) Control Valves | 5 |
| | | (ii) Pneumatic Control | 5 |
| | | (iii) Automatic control | |

OR

- | | | | |
|---|---|--|---|
| X | a | Describe the following | |
| | | (i) Distributed Control Systems | 5 |
| | | (ii) Supervisory and Direct control System | 5 |
| | | (iii) Digital computer | 5 |